KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION · CSE

# DIABETES-RELATED NAFLD / MASLD CLINICAL DATASET

Explainable AI Early Diagnostic Benchmark Dataset (5,847 Records · 28 Features)

## 1. Research & System Metadata

• Project Title: Personalized AI-Based Healthcare System for Early Diagnosis of Diabetes-Related NAFLD
• Project Guide: Dr. R. Murugeswari, Associate Professor, Department of Computer Science & Engineering
• Student Investigators: Ch. Pardha Saradhi (9924051026), Ch. Sai Dheeraj (9924051028), G. Shiva Shankara Vara Prasad (9924051008), B. Vinay (9924051023)
• Clinical Objective: Early non-invasive detection of Non-Alcoholic Fatty Liver Disease (NAFLD/MASLD) in Type-2 Diabetes patients prior to irreversible hepatic fibrosis.

## 2. Dataset Technical Architecture (Slide 12 Specification)

| Total Cohort Records | Clinical Features | Classification Target | Train / Test Split | Missing Imputation |
|---|---|---|---|---|
| 5,847 Patients | 28 Variables | Binary (NAFLD / Normal) | 80% (4,677) / 20% (1,170) | KNN / Iterative Imputer |

## 3. Feature Domain Taxonomy (28 Parameters)

**Demographic Baseline**

Age, Biological Gender, BMI (Body Mass Index), Ethnicity / Region

**4 Features**

**Clinical & Vascular History**

Systolic Blood Pressure, Diastolic Blood Pressure, Diabetes Duration (years), Family History of T2D, Family History of Hepatic Disease, Waist-to-Hip Ratio

**6 Features**

**Laboratory Biomarkers**

Fasting Blood Glucose, HbA1c Glycemia, ALT / SGPT, AST / SGOT, AST/ALT Ratio, Platelet Count, FIB-4 Score, APRI Score, Triglycerides, HDL, LDL, Total Cholesterol, Alkaline Phosphatase

**13 Features**

**Lifestyle & Environmental**

Healthy Eating Index (HEI-2020), Weekly Exercise Duration (hrs), Sleep Quality Score, Smoking Status, Alcohol Consumption Frequency

**5 Features**

## 4. Machine Learning & Explainable AI Benchmarks (Slide 6 & 13)

| Algorithm | Working Paradigm | Accuracy | Sensitivity | Specificity | AUC-ROC |
|---|---|---|---|---|---|
| CatBoost Classifier | Ordered Gradient Boosting (Categorical Focus) | 95.6% | 96.1% | 94.8% | 0.978 |
| Artificial Neural Net (ANN) | Deep Multi-Layer Perceptron (ReLU, Adam) | 94.8% | 95.0% | 94.4% | 0.971 |
| Random Forest Ensemble | Bagged Decision Trees (Gini Impurity) | 94.2% | 93.8% | 94.7% | 0.965 |
| Support Vector Machine | Radial Basis Function (RBF Kernel) | 92.8% | 91.9% | 93.5% | 0.948 |
| LIME / SHAP Explainability | Local Linear Surrogates + Shapley Attributions | — | — | — | Verified XAI |

## 5. Existing Healthcare System vs. Proposed Deep Learning Platform

Quantitative comparative capability scores based on the research presentation (Feature-Coverage Scale: 0 to 5):

| Evaluation Capability Criterion | Existing Standard Clinical Workflow | Our Proposed Deep-Learning Platform | Clinical Benefit |
|---|---|---|---|
| Automated Risk Prediction | 2 / 5 (Manual blood tests) | **5 / 5 (Real-time AI engine)** | Immediate stratification at outpatient visit |
| Complex Pattern Detection | 2 / 5 (Isolated lab ranges) | **5 / 5 (Deep non-linear ANN)** | Detects silent hepatocyte injury early |
| Personalized Risk Assessment | 2 / 5 (Generalized thresholds) | **5 / 5 (Patient-specific vector)** | Individualized lifestyle & clinical pathway |
| Automated Data Analysis | 2 / 5 (Fragmented paper/ EHR) | **5 / 5 (Continuous IoT ingestion)** | Zero-latency ingestion from CGM & wearables |
| Prediction Explanations | 2 / 5 (Black-box doctor note) | **4 / 5 (LIME local explanations)** | High clinician confidence with XAI factors |
| Patient Monitoring Support | 2 / 5 (Infrequent annual visit) | **5 / 5 (Continuous smart telemetry)** | Prevents irreversible cirrhosis progression |

## 6. Validated Sample Patient Cohort Records

| PID | Name | Age/Sex | BMI | Glucose | HbA1c | ALT/AST | FIB-4 | Risk Tier |
|---|---|---|---|---|---|---|---|---|
| PID-8921 | Ramesh Kumar | 56M | 32.4 | 168 | 8.8% | 64/52 | 1.92 | **High Risk** |
| PID-8922 | Priya Lakshmi | 49F | 29.1 | 145 | 7.6% | 48/38 | 1.25 | **Moderate Risk** |
| PID-8923 | Mohammed Rafiq | 63M | 34.2 | 184 | 9.2% | 72/58 | 2.46 | **High Risk** |
| PID-8924 | Ananya Subramanian | 44F | 27.8 | 132 | 7.2% | 42/35 | 1.01 | **Moderate Risk** |
| PID-8925 | Dr. Vignesh Murugan | 38M | 23.2 | 98 | 5.8% | 22/20 | 0.60 | **Low Risk** |
| PID-8926 | Kavitha Ramachandran | 41F | 24.1 | 104 | 6.1% | 26/24 | 0.76 | **Low Risk** |